Reg. No									

St. Joseph's College of Arts & Science (Autonomous)

St. Joseph's College Road, Cuddalore – 607001

ASMT101Q - MATHEMATICAL STATISTICS - I

Time : 3 hrs **Max Marks :75**

SECTION – A (10x2=20)

Answer ALL the Questions

1. Mention the types of bar diagram.
2. Define arithmetic mean.
3. Define probability.
4. Define independent events.
5. Define cumulative distribution function.
6. When do you say the given function is said to be probability density function?
7. Define expectation of a random variable.
8. Define Chebychevs inequality.
9. Define correlation.
10. Write down the regression equation of y on x.

SECTION – B (5x5=25)

Answer ALL the Questions

11. a. Draw the frequency histogram and polygon for the following data.

Marks	0 -10	10 - 20	20 - 30	30 - 40	40 -50	50 - 60
No. of Students	5	10	15	30	5	5

(OR)

b. What do you understand by measures of central values and mention the various measures of central values?

12. a. State and prove addition theorem on probability.

(OR)

b. Prove that, if A and B are independent events then A' and B' are also independent events.

13. a. A random variableX has the following probability distribution.

x	0	1	2	3	4	5	6	7	8
P(x)	a	3a	5a	7a	9a	11a	13a	15a	17a

i) Determine the value of a.

ii) Find P(X<3).

(OR)

b. The joint pdf of a two dimensional random variable (x.y) is given by

f(x,y) = 2 ; 0<x<1, 0<y<x

= 0, elsewhere

14. a. State and prove the addition theorem on expectation.

(OR)

b. What is cumulant? State the relationship between the moments and cumulants.

15. a. Explain Spearmanns rank correlation.

(OR)

b. State the different methods of studying correlation and explain any one of the method.

SECTION – C (3x10 = 30)

Answer any **THREE** questions.

16. Explain the diagrammatic representation of statistical data.
17. State and prove Bayes theorem.
18. The joint density of X and Yis given by f(x,y)= 4xy exp(-(x^2+y^2)) ; x≥0, y≥ Test whether X and Y are independent. For the above joint density, find the conditional density of X given Y = y.
19. Obtain the moment generating function of binomial distribution and deduce its mean and variance.
20. Obtain the regression equation of Y on X for the following data.

X:	61	72	73	63	84	80	66	76
Y:	40	52	49	43	61	58	42	58

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